# Predicting NBA Playoff Success from Team Stats

A regression-based playoff round predictor

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**Presented by**

Judah Del Rio

Shakir Islam

David Tagoe

## Our Question

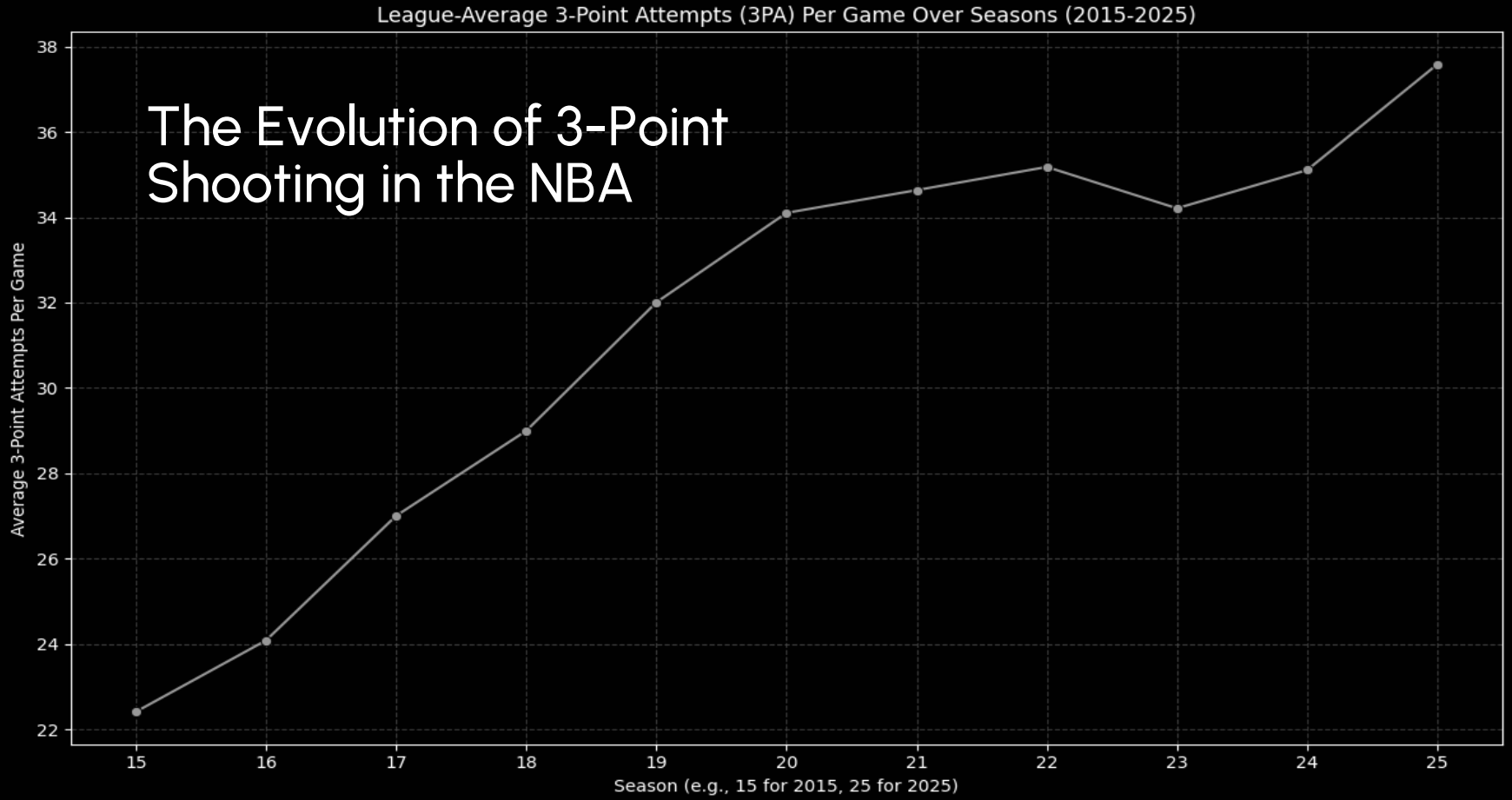
# Which NBA statistic is the best predictor for playoff success?

There are so many counting stats. Which ones impact winning the most?

Our hypothesis: Among standard counting stats, a team's 3-point percentage is the greatest determining factor for its playoff success.

| Rk | Team | G | MP | FG | FGA | FG% | 3P | 3PA | 3P% | 2P | 2PA | 2P% | FT | FTA | FT% | ORB | DRB | TRB | AST | STL | BLK | TOV | PF | PTS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Atlanta Hawks | 82 | 241.2 | 43.4 | 91.8 | .472 | 13.5 | 37.7 | .358 | 29.9 | 54.1 | .552 | 18.0 | 23.2 | .775 | 11.9 | 32.6 | 44.5 | 29.6 | 9.7 | 5.1 | 15.5 | 19.1 | 118.2 |
| 2 | Boston Celtics* | 82 | 241.8 | 41.6 | 90.0 | .462 | 17.8 | 48.2 | .368 | 23.9 | 41.8 | .571 | 15.3 | 19.1 | .799 | 11.4 | 33.9 | 45.3 | 26.1 | 7.2 | 5.5 | 11.9 | 15.9 | 116.3 |
| 3 | Brooklyn Nets | 82 | 241.2 | 37.6 | 86.1 | .437 | 13.6 | 39.4 | .344 | 24.1 | 46.7 | .515 | 16.3 | 20.7 | .787 | 10.9 | 30.4 | 41.3 | 25.2 | 7.8 | 4.3 | 15.2 | 20.7 | 105.1 |
| 4 | Charlotte Hornets | 82 | 240.9 | 38.3 | 89.1 | .430 | 13.0 | 38.3 | .339 | 25.3 | 50.8 | .499 | 15.5 | 19.8 | .783 | 12.2 | 33.0 | 45.2 | 24.3 | 7.4 | 4.5 | 15.5 | 18.7 | 105.1 |
| 5 | Chicago Bulls | 82 | 240.9 | 43.2 | 92.0 | .470 | 15.4 | 42.0 | .367 | 27.8 | 50.0 | .556 | 16.0 | 19.7 | .809 | 10.1 | 35.8 | 45.9 | 29.1 | 7.6 | 4.7 | 14.7 | 17.6 | 117.8 |
| 6 | Cleveland Cavaliers* | 82 | 240.9 | 44.5 | 90.8 | .491 | 15.9 | 41.5 | .383 | 28.6 | 49.3 | .581 | 17.0 | 21.9 | .776 | 11.2 | 34.2 | 45.4 | 28.1 | 8.2 | 4.3 | 13.2 | 18.1 | 121.9 |
| 7 | Dallas Mavericks | 82 | 240.6 | 42.0 | 87.7 | .479 | 12.4 | 34.2 | .364 | 29.5 | 53.6 | .552 | 17.8 | 23.1 | .770 | 10.1 | 33.0 | 43.0 | 25.2 | 7.8 | 5.4 | 14.0 | 17.8 | 114.2 |
| 8 | Denver Nuggets* | 82 | 242.1 | 45.4 | 89.8 | .506 | 12.0 | 31.9 | .376 | 33.4 | 57.8 | .578 | 17.9 | 23.3 | .770 | 11.2 | 34.5 | 45.7 | 31.0 | 8.0 | 4.9 | 14.3 | 17.6 | 120.8 |
| 9 | Detroit Pistons* | 82 | 241.5 | 42.7 | 89.8 | .476 | 12.8 | 35.4 | .362 | 29.9 | 54.4 | .550 | 17.2 | 22.3 | .774 | 11.4 | 33.4 | 44.8 | 26.4 | 8.0 | 5.3 | 14.9 | 20.7 | 115.5 |
| 10 | Golden State Warriors* | 82 | 240.6 | 40.8 | 90.4 | .451 | 15.4 | 42.4 | .364 | 25.3 | 48.0 | .528 | 16.9 | 22.1 | .764 | 12.5 | 32.8 | 45.4 | 29.1 | 9.4 | 4.8 | 14.0 | 19.2 | 113.8 |
| 11 | Houston Rockets* | 82 | 241.2 | 42.5 | 93.4 | .455 | 12.7 | 35.8 | .353 | 29.8 | 57.6 | .518 | 16.7 | 22.6 | .738 | 14.6 | 33.8 | 48.5 | 23.3 | 8.4 | 5.0 | 13.9 | 19.0 | 114.3 |
| 12 | Indiana Pacers* | 82 | 242.1 | 43.6 | 89.3 | .488 | 13.2 | 35.8 | .368 | 30.4 | 53.6 | .568 | 17.0 | 21.6 | .789 | 9.2 | 32.7 | 41.8 | 29.2 | 8.5 | 5.5 | 13.2 | 18.7 | 117.4 |
| 13 | Los Angeles Clippers* | 82 | 241.5 | 41.6 | 86.3 | .482 | 12.5 | 33.4 | .373 | 29.1 | 52.9 | .550 | 17.3 | 21.7 | .797 | 10.3 | 33.5 | 43.8 | 25.2 | 9.4 | 4.5 | 14.9 | 18.3 | 112.9 |
| 14 | Los Angeles Lakers* | 82 | 240.6 | 40.9 | 85.5 | .479 | 13.3 | 36.4 | .366 | 27.6 | 49.1 | .562 | 18.2 | 23.2 | .785 | 9.7 | 32.8 | 42.4 | 26.0 | 7.7 | 4.5 | 14.0 | 17.3 | 113.4 |
| 15 | Memphis Grizzlies* | 82 | 240.3 | 44.8 | 93.3 | .479 | 13.9 | 37.9 | .367 | 30.8 | 55.5 | .556 | 18.3 | 23.3 | .786 | 12.9 | 34.4 | 47.3 | 28.4 | 8.9 | 5.6 | 15.7 | 20.9 | 121.7 |
| 16 | Miami Heat* | 82 | 243.0 | 40.5 | 87.0 | .465 | 13.7 | 37.3 | .367 | 26.8 | 49.7 | .538 | 16.0 | 20.2 | .791 | 9.7 | 33.7 | 43.4 | 26.4 | 8.1 | 3.8 | 13.6 | 15.5 | 110.6 |
| 17 | Milwaukee Bucks* | 82 | 240.9 | 42.0 | 86.4 | .486 | 14.2 | 36.6 | .387 | 27.9 | 49.7 | .560 | 17.3 | 23.1 | .750 | 8.2 | 35.1 | 43.4 | 25.5 | 7.3 | 4.7 | 13.4 | 17.9 | 115.5 |
| 18 | Minnesota Timberwolves* | 82 | 241.8 | 41.0 | 87.6 | .468 | 15.0 | 39.9 | .377 | 26.0 | 47.7 | .545 | 17.2 | 21.8 | .789 | 11.1 | 33.2 | 44.3 | 26.1 | 8.0 | 5.0 | 14.5 | 18.3 | 114.3 |
| 19 | New Orleans Pelicans | 82 | 240.9 | 40.6 | 89.9 | .452 | 12.0 | 34.6 | .347 | 28.6 | 55.2 | .518 | 16.5 | 21.9 | .754 | 12.1 | 31.5 | 43.6 | 25.8 | 8.5 | 5.2 | 14.6 | 18.3 | 109.8 |
| 20 | New York Knicks* | 82 | 242.4 | 43.3 | 89.2 | .486 | 12.6 | 34.1 | .369 | 30.8 | 55.1 | .558 | 16.5 | 20.7 | .800 | 10.9 | 31.8 | 42.6 | 27.5 | 8.2 | 4.0 | 13.3 | 17.2 | 115.8 |
| 21 | Oklahoma City Thunder* | 82 | 240.3 | 44.6 | 92.7 | .482 | 14.5 | 38.8 | .374 | 30.1 | 53.9 | .559 | 16.7 | 20.4 | .819 | 10.6 | 34.2 | 44.8 | 26.9 | 10.3 | 5.7 | 11.7 | 19.9 | 120.5 |
| 22 | Orlando Magic* | 82 | 240.6 | 38.2 | 85.8 | .445 | 11.2 | 35.3 | .318 | 26.9 | 50.5 | .533 | 17.9 | 23.0 | .775 | 11.1 | 30.7 | 41.8 | 23.0 | 8.9 | 6.0 | 14.2 | 20.1 | 105.4 |
| 23 | Philadelphia 76ers | 82 | 241.5 | 39.7 | 87.4 | .454 | 12.7 | 37.2 | .341 | 27.0 | 50.2 | .537 | 17.5 | 22.5 | .780 | 10.4 | 29.4 | 39.8 | 23.2 | 9.2 | 4.5 | 13.6 | 19.0 | 109.6 |
| 24 | Phoenix Suns | 82 | 241.8 | 41.2 | 86.3 | .478 | 14.3 | 38.0 | .378 | 26.9 | 48.3 | .556 | 16.8 | 20.8 | .810 | 9.5 | 32.9 | 42.5 | 27.8 | 7.2 | 4.8 | 14.1 | 17.9 | 113.6 |
| 25 | Portland Trail Blazers | 82 | 241.2 | 40.5 | 90.2 | .450 | 12.9 | 37.7 | .342 | 27.7 | 52.5 | .527 | 16.9 | 22.2 | .762 | 13.4 | 31.4 | 44.8 | 23.8 | 8.3 | 5.3 | 16.0 | 19.0 | 110.9 |
| 26 | Sacramento Kings | 82 | 242.4 | 43.0 | 90.1 | .478 | 12.6 | 35.2 | .357 | 30.5 | 54.9 | .555 | 17.1 | 21.2 | .806 | 11.0 | 33.2 | 44.2 | 26.5 | 7.6 | 4.4 | 13.3 | 18.9 | 115.7 |
| 27 | San Antonio Spurs | 82 | 240.6 | 41.8 | 89.8 | .465 | 14.1 | 39.6 | .357 | 27.7 | 50.2 | .551 | 16.2 | 21.0 | .774 | 10.9 | 32.7 | 43.7 | 28.6 | 8.2 | 5.5 | 13.9 | 16.6 | 113.9 |
| 28 | Toronto Raptors | 82 | 241.5 | 41.6 | 91.0 | .458 | 11.8 | 34.0 | .348 | 29.8 | 57.0 | .523 | 15.8 | 21.1 | .748 | 12.6 | 32.5 | 45.1 | 28.5 | 8.1 | 4.2 | 15.3 | 21.2 | 110.9 |
| 29 | Utah Jazz | 82 | 241.5 | 40.3 | 88.7 | .454 | 14.0 | 39.8 | .350 | 26.4 | 48.8 | .540 | 17.3 | 22.4 | .774 | 12.0 | 33.4 | 45.4 | 25.5 | 6.8 | 4.4 | 17.2 | 18.9 | 111.9 |
| 30 | Washington Wizards | 82 | 240.6 | 39.4 | 89.9 | .439 | 13.1 | 39.1 | .335 | 26.3 | 50.7 | .519 | 16.0 | 20.6 | .778 | 10.6 | 33.1 | 43.7 | 25.1 | 7.6 | 5.1 | 15.6 | 19.7 | 108.0 |
| | **League Average** | **82** | **241.3** | **41.7** | **89.2** | **.467** | **13.5** | **37.6** | **.360** | **28.2** | **51.7** | **.545** | **16.9** | **21.7** | **.780** | **11.1** | **33.0** | **44.1** | **26.5** | **8.2** | **4.9** | **14.3** | **18.6** | **113.8** |

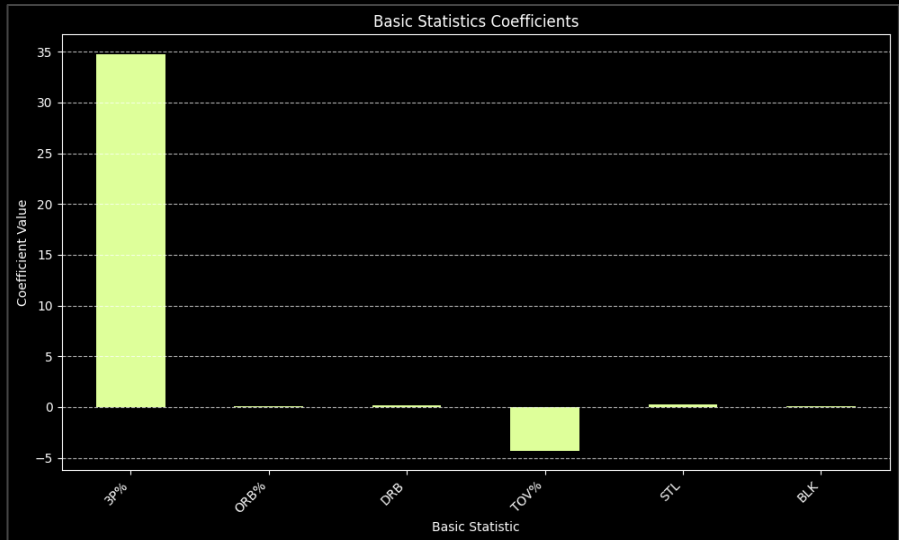
# The Evolution of 3-Point Shooting in the NBA



# The Method

- 11 Years ago, Steph Curry and his Warriors changed the way NBA basketball is played forever.
- We tracked every relevant counting stat in those years for all 30 teams – 330 seasons total.
- Playoff success tracked on a 0-5 scale (0 missed the playoffs, 1 lost in the first round, 5 won the championship, etc.)

| Team | Season | Playoff_Rd | TS% | 3P% | ORB% | DRB | TOV% | STL | BLK |
|---|---|---|---|---|---|---|---|---|---|
| Atlanta Hawks | 25 | 0 | 0.579366324 | 0.358 | 26.3 | 32.6 | 0.131905913 | 9.7 | 5.1 |
| Boston Celtics | 25 | 2 | 0.590931263 | 0.368 | 25.7 | 33.9 | 0.107883667 | 7.2 | 5.5 |
| Brooklyn Nets | 25 | 0 | 0.551949416 | 0.344 | 24.5 | 30.4 | 0.137671183 | 7.8 | 4.3 |
| Charlotte Hornets | 25 | 0 | 0.537255143 | 0.339 | 26.5 | 33 | 0.136790455 | 7.4 | 4.5 |
| Chicago Bulls | 25 | 0 | 0.585091588 | 0.367 | 22.3 | 35.8 | 0.127418348 | 7.6 | 4.7 |
| Cleveland Cavaliers | 25 | 2 | 0.606854116 | 0.383 | 25.9 | 34.2 | 0.116160372 | 8.2 | 4.3 |
| Dallas Mavericks | 25 | 0 | 0.583462765 | 0.364 | 23.3 | 33 | 0.12515197 | 7.8 | 5.4 |
| Denver Nuggets | 25 | 2 | 0.603686083 | 0.376 | 26.7 | 34.5 | 0.12505247 | 8 | 4.9 |
| Detroit Pistons | 25 | 1 | 0.579749428 | 0.362 | 26.2 | 33.4 | 0.130117368 | 8 | 5.3 |




| *Regression Statistics* | |
|---|---|
| Multiple R | 0.575923747 |
| R Square | 0.331688163 |
| Adjusted R Square | 0.319273701 |
| Standard Error | 1.078295366 |
| Observations | 330 |

## Basic Stats Regression – 3PT% Wins!

Our model clearly indicates that 3P% is the strongest positive driver for playoff success with a P-value of 2.37E-17, while TOV%, also a significant factor in the opposite direction, had a P-value of 0.5.

Other statistics like ORB%, DRB, STL and BLK contribute to a lesser extent to the predicted outcome.

# That Was Easy...
# But What About Some Advanced Stats?




Second hypothesis: True shooting will outperform 3 point percentage as a predictor for playoff success by a significant margin.

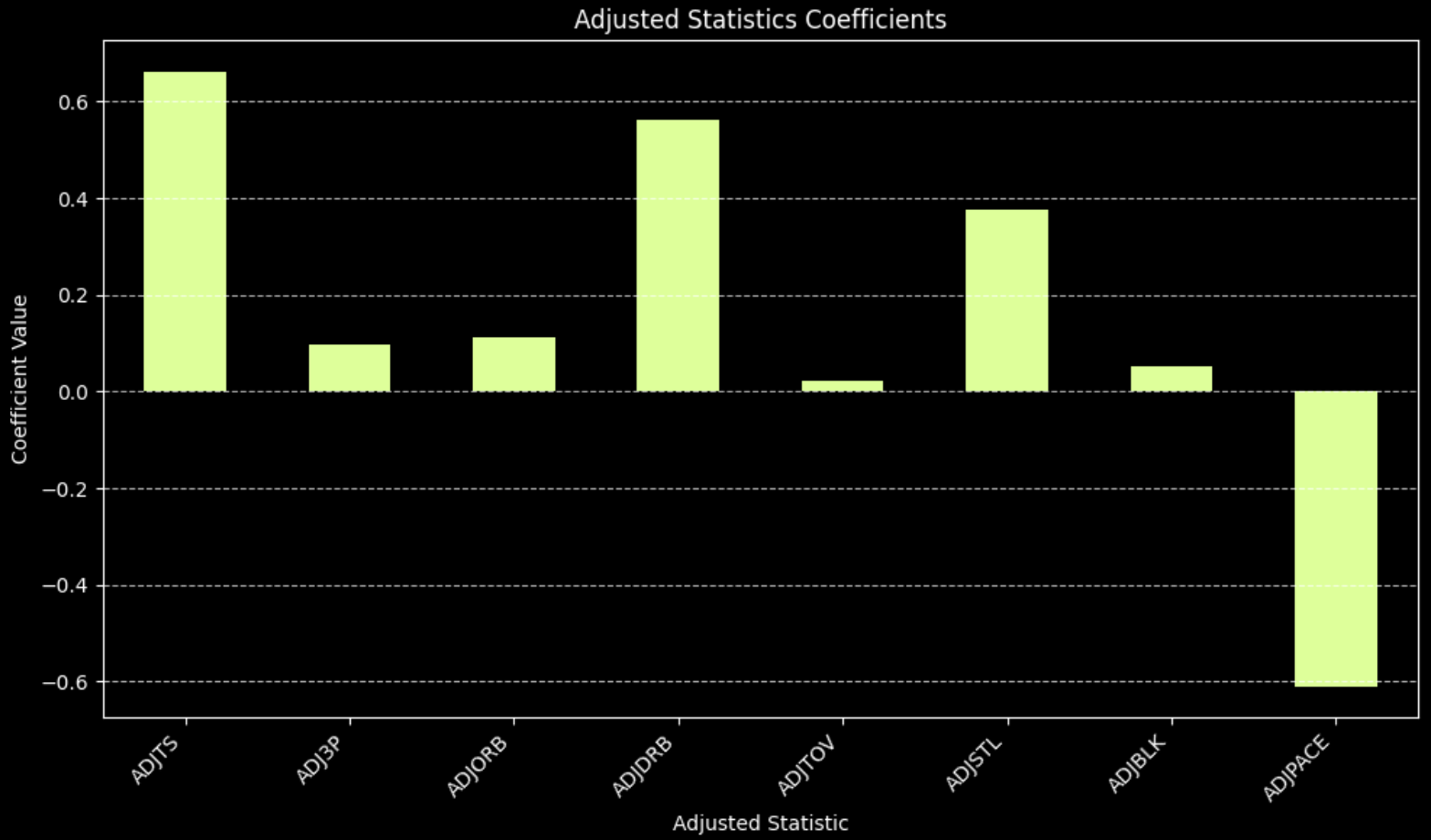
True shooting percentage is simple:
PTS / (2 * (FGA + 0.44 * FTA))

Pace is a bit more complex:
48 * ((Team Possessions + Opponents' Possessions) / (2 * (Team's Minutes Played/ 5)))

In other words, true shooting is how efficient an offense is overall, with every type of shot included, and pace is a measure of how fast a team plays in comparison to other teams.

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.684594177 |
| R Square | 0.468669187 |
| Adjusted R Square | 0.455427298 |
| Standard Error | 0.964449313 |
| Observations | 330 |

TS% and Pace strengthen our regression by a significant amount, a good sign for the integrity of our second hypothesis.


Adjusted Statistics Coefficients
Coefficient Value
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
ADJTS
ADJ3P
ADJORB
ADJDRB
ADJTOV
ADJSTL
ADJBLK
ADJPACE
Adjusted Statistic

# TS% vs 3PT%

- True shooting and 3PT% correlate heavily.
- Not enough to sound alarms.
- True shooting measures what 3PT% measures, but better.



# The NBA's Ultimate Advanced Stat




Final hypothesis: Net rating will outperform not only 3 point percentage, but also true shooting as a predictor for playoff success by a significant margin.

**Net Rating measures** a given team's point differential per 100 possessions with the formula:

$$100 \times \left(\frac{\text{Points}}{\text{POSS}}\right) - 100 \times \left(\frac{\text{Opp Points}}{\text{Opp POSS}}\right)$$

It serves as the best overall predictive statistic in the NBA since it is so all-encompassing.

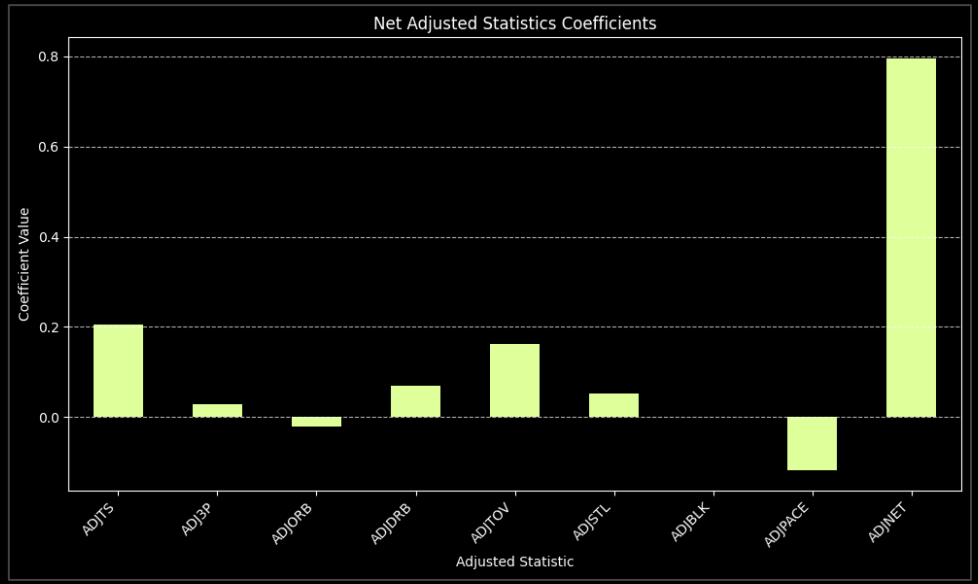
## Net Rating Dominates the Regression

Net rating allows our regression to predict nba playoff outcomes with over 50 percent accuracy!

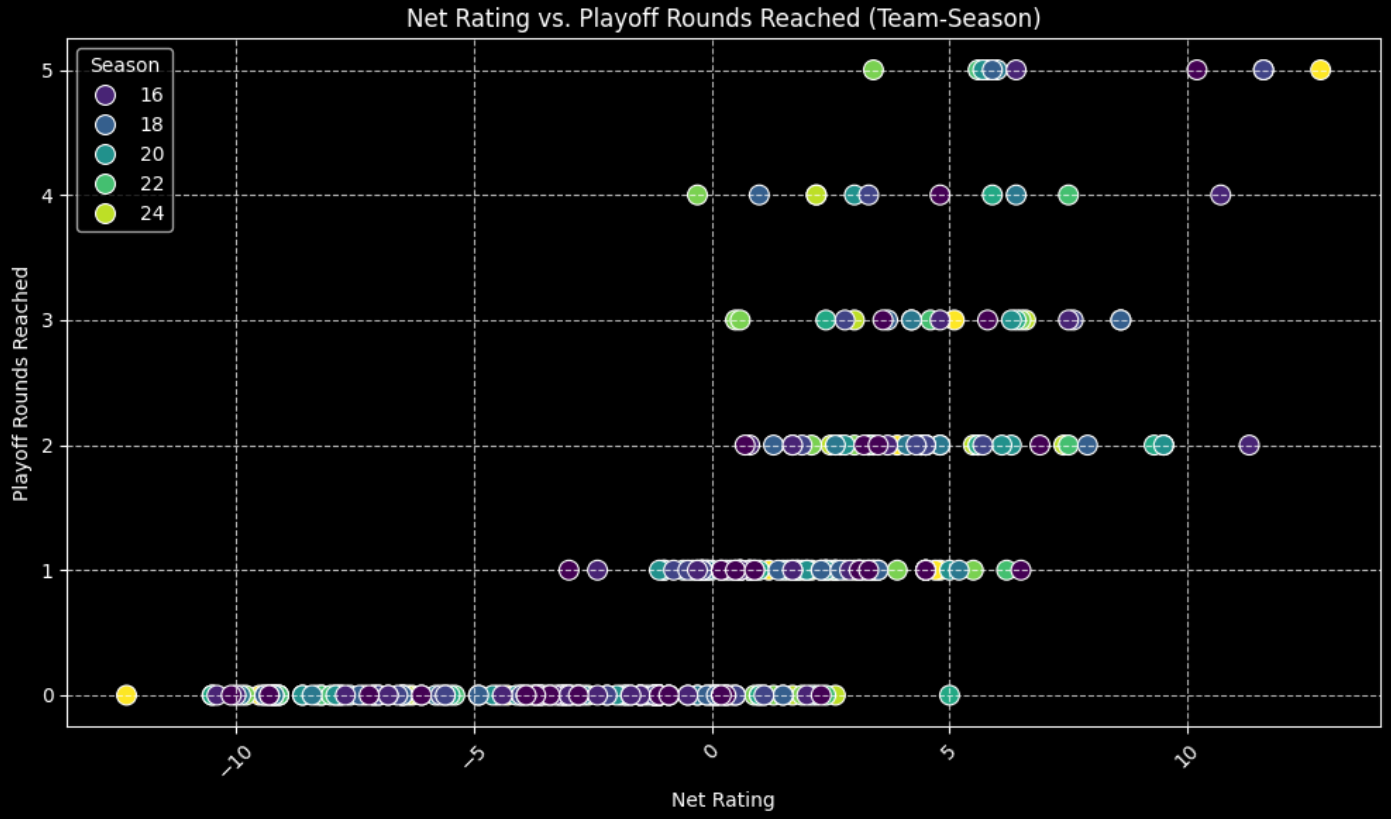
For every increase in one standard deviation of net rating, teams can expect to increase their playoff success by 0.8 whole rounds. It even has a P-value of 5.22E-09!

True shooting matters, but pales in comparison.

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.722816129 |
| R Square | 0.522463156 |
| Adjusted R Square | 0.509032432 |
| Standard Error | 0.915752002 |
| Observations | 330 |



# Net Rating Dominates the Regression



# Using the coefficients from our final regression, we derived:

```
= 1.012121212
  + 0.206266019 * ADJTS
  + 0.095893981 * ADJ3P
  - 0.020069357 * ADJORB
  + 0.086878116 * ADJDRB
  + 0.161588871 * ADJTOV
  + 0.051497539 * ADJSTL
  + 0.006564867 * ADJBLK
  - 0.116857749 * ADJPACE
  + 0.796384695 * ADJNET
```

This gives us an ugly decimal, which we order by rank within each season (1-30 for each team for all 11 years), and then assign a playoff round according to their rank among teams.

Our Playoff Predictor in Action!

With over 50 percent accuracy, our formula can predict any given NBA team's playoff performance!

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.722816129 |
| R Square | 0.522463156 |
| Adjusted R Square | 0.509032432 |
| Standard Error | 0.915752002 |
| Observations | 330 |

| Team | Season | Playoff_Rd | Final Predicted Playoff Round |
|---|---|---|---|
| Atlanta Hawks | 25 | 0 | 0 |
| Boston Celtics | 25 | 2 | 3 |
| Brooklyn Nets | 25 | 0 | 0 |
| Charlotte Hornets | 25 | 0 | 0 |
| Chicago Bulls | 25 | 0 | 0 |
| Cleveland Cavaliers | 25 | 2 | 4 |
| Dallas Mavericks | 25 | 0 | 0 |
| Denver Nuggets | 25 | 2 | 2 |
| Detroit Pistons | 25 | 1 | 1 |
| Golden State Warriors | 25 | 2 | 1 |
| Houston Rockets | 25 | 1 | 1 |
| Indiana Pacers | 25 | 4 | 1 |
| Los Angeles Clippers | 25 | 1 | 3 |
| Los Angeles Lakers | 25 | 1 | 1 |
| Memphis Grizzlies | 25 | 1 | 2 |
| Miami Heat | 25 | 1 | 1 |
| Milwaukee Bucks | 25 | 1 | 2 |
| Minnesota Timberwolves | 25 | 3 | 2 |
| New Orleans Pelicans | 25 | 0 | 0 |
| New York Knicks | 25 | 3 | 1 |
| Oklahoma City Thunder | 25 | 5 | 5 |
| Orlando Magic | 25 | 1 | 0 |
| Philadelphia 76ers | 25 | 0 | 0 |
| Phoenix Suns | 25 | 0 | 0 |
| Portland Trail Blazers | 25 | 0 | 0 |
| Sacramento Kings | 25 | 0 | 1 |
| San Antonio Spurs | 25 | 0 | 0 |
| Toronto Raptors | 25 | 0 | 0 |
| Utah Jazz | 25 | 0 | 0 |
| Washington Wizards | 25 | 0 | 0 |

# Thank you!

